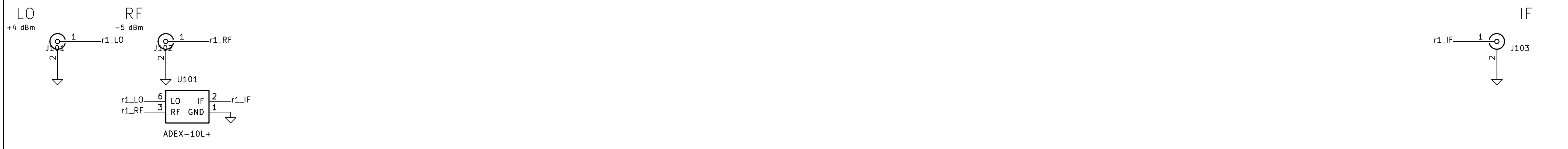
r1 diode mixer, ADEX-10L+

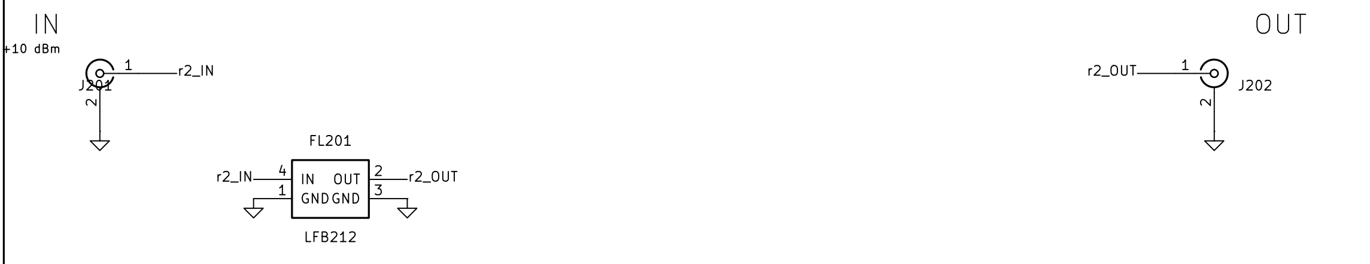
cells (0,0), (1,0)


LO
+4 dBm
RF
−5 dBm
IF
J101
J102
J103
U101
ADEX-10L+
r1_LO
r1_RF
r1_IF

EXPECT: about 7 dB of conversion loss from 10 MHz to 1 GHz with +4 dBm of LO (r10 at its top power), IF from DC to 800 MHz, 60 dB L-R isolation
Mini-Circuits ADEX-10L+: two wideband baluns and a ring of four Schottky diodes (drawn beside it on the silk). The LO (+4 dBm, +1 to +7) turns the diodes on in pairs, so the RF is multiplied by +-1 at the LO rate: the IF is RF x square wave, the sum and difference plus odd LO harmonics. Passive and DC-coupled at IF (a phase detector when RF and LO share a frequency); 1 dB compression at +1 dBm of RF, IP3 +16 dBm. Compare r7/r8 (ADL5350) and AMOD-A's a9 (MC1496).

r2 2.45 GHz filter

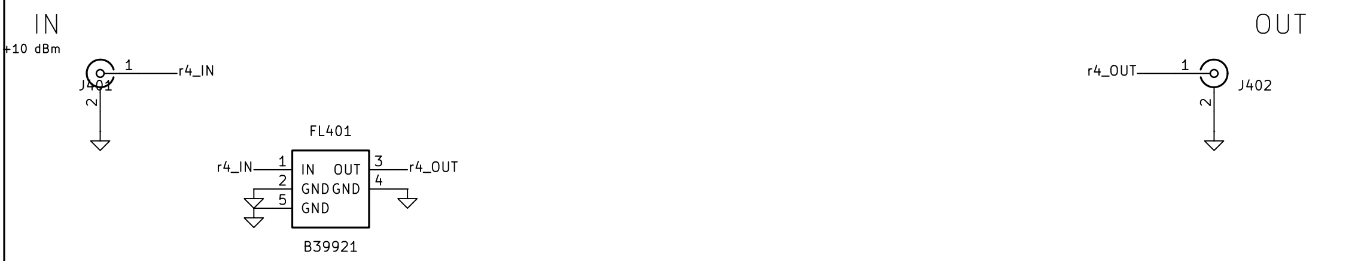
cell (2,0)


IN
+10 dBm
OUT
J201
J202
FL201
LFB212
r2_IN
r2_OUT

EXPECT: 1.7 dB of loss from 2.4 to 2.5 GHz, 30 dB or more below 2.2 GHz and above 2.7 GHz (the datasheet); 50 ohm in and out
Murata LFB212G45SG8A127, a 2012-size LTCC band-pass. The line to each jack is 50 ohm coplanar waveguide (0.29 mm wide, 0.25 to the ground pour) with a via fence; the jack's centre pin has no plane under it (a 2.2 mm void).
Sweep it with r10 or r13 into a power meter, or with a DAC harmonic comb and the ADC.

r4 915 MHz filter

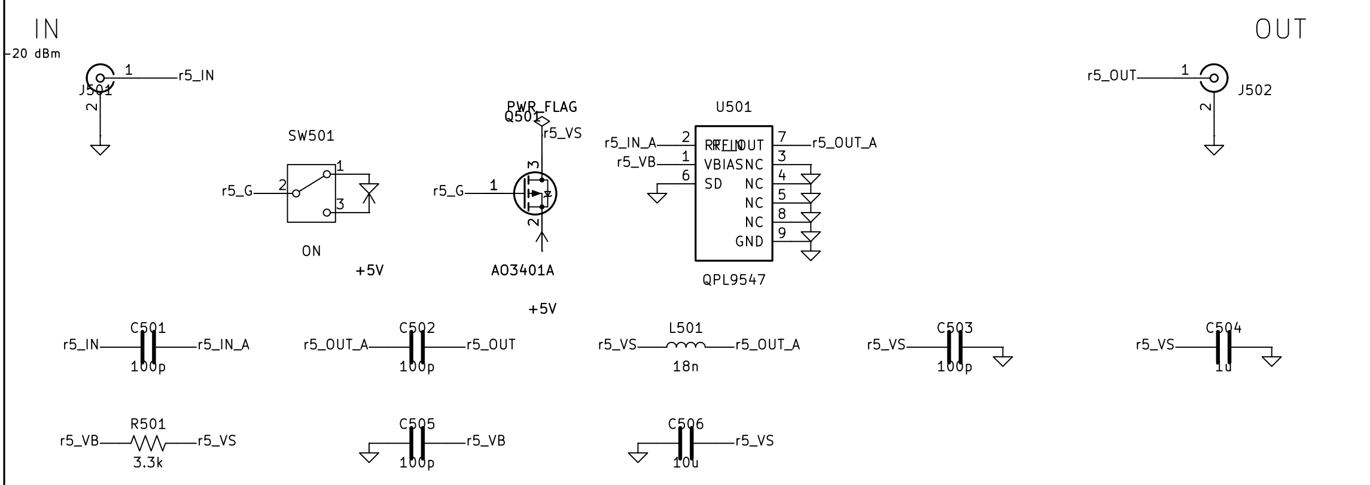
cell (3,0)


IN
+10 dBm
OUT
J401
J402
FL401
B39921
r4_IN
r4_OUT

EXPECT: 1.1 dB of loss from 902 to 928 MHz, 40 dB or more 30 MHz outside the band (the datasheet); 50 ohm in and out
RF360 B39921B2672P810: a surface-acoustic-wave filter for the 915 MHz ISM band, 26 MHz wide, in a 1411 package; its skirts are much steeper than the LTCC's. Lines, fences and launch voids as on r2.

r5 LNA, QPL9547

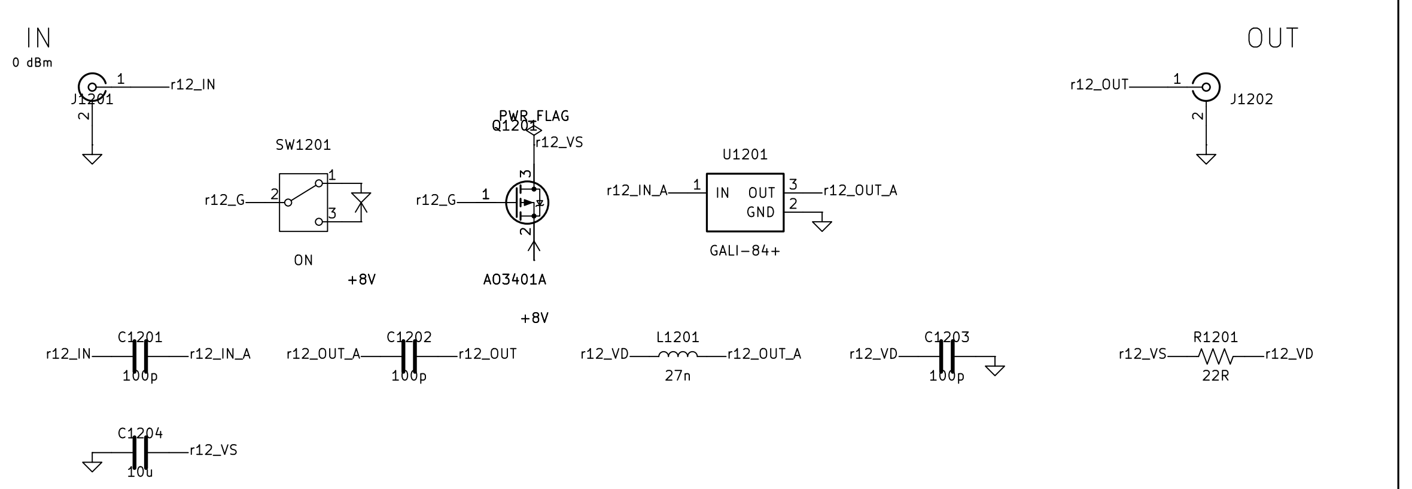
cell (0,1)


IN
−20 dBm
OUT
J501
J502
SW501
ON
PWR_FLAG
Q501
AO3401A
U501
QPL9547
+5V
r5_IN
r5_OUT
r5_VS
r5_VB
r5_G
L501
18n
R501
3.3k
100p
10u

EXPECT: 19.5 dB of gain and a 0.3-0.5 dB noise figure from 0.5 to 3 GHz (datasheet: 0.3 dB at 1.9 GHz, OIP3 +39 dBm); 65 mA from 5 V
Qorvo QPL9547: a GaAs pHEMT LNA, internally matched to 50 ohm, 0.1-6 GHz. It sets the noise figure of every receiver here, so put it first (before the filter) unless a strong neighbour overloads it; +39 dBm OIP3 makes that rare.
The 18 nH wire-wound choke (Qorvo's value) feeds 5 V into the output pin; 3.3 k from 5 V to Vbias sets 65 mA.
Never cable a transmitter into it without 30 dB of attenuation (22 dBm absolute maximum at the input).

r12 gain block, GALI-84+

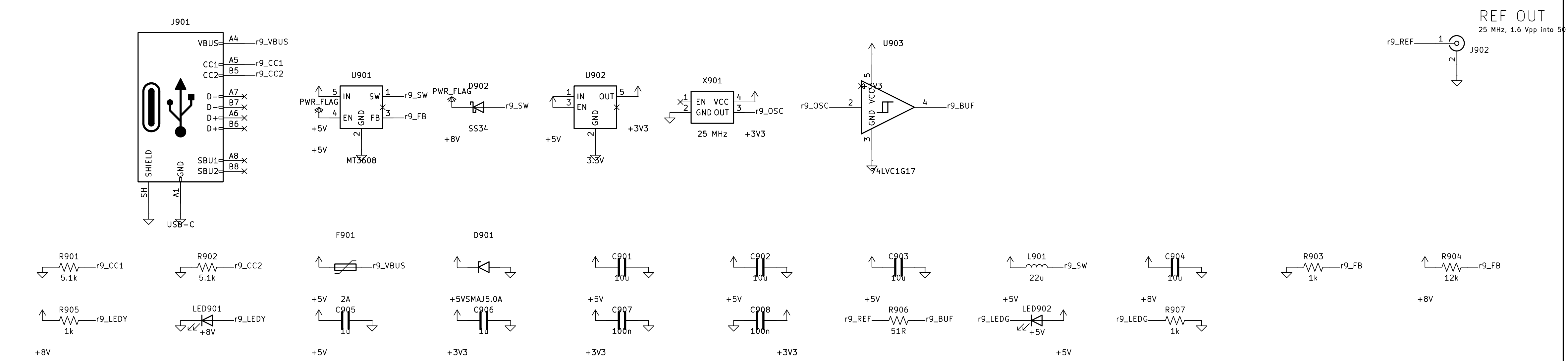
cell (1,1)


IN
0 dBm
OUT
J1201
J1202
SW1201
ON
PWR_FLAG
Q1201
AO3401A
U1201
GALI-84+
+8V
r12_IN
r12_OUT
r12_VS
r12_VD
L1201
27n
R1201
22R
100p
10u

EXPECT: 20 dB of gain from 50 MHz to 2.5 GHz (datasheet: 21.9 dB at 1 GHz, 1 dB compression at +20 dBm out); 97 mA from 7.8 V
Mini-Circuits GALI-84+: a Darlington MMIC that sets its own current; the 22 ohm from the 7.8 V rail drops to its 5.8 V device voltage (0.2 W in the 2512). The 27 nH choke (415 ohm at 2.45 GHz, 155 at 915 MHz) feeds the output pin; 100 pF blocks keep the DC off the jacks. Keep the input under 0 dBm for linear gain. The slide switch and P-FET gate the tile's 7.8 V as on AMOD-A.

r9 power, USB-C 5 V in: +5, +7.8 V; 25 MHz reference

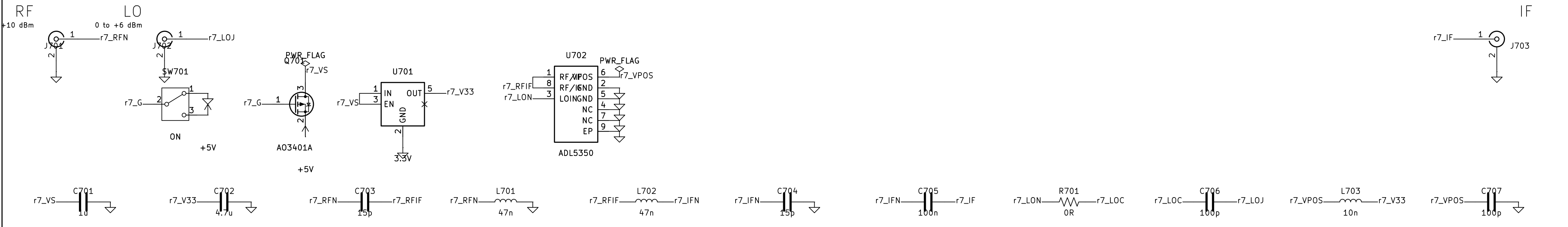
cells (2,1), (3,1)


REF OUT
25 MHz, 1.6 Vpp into 50 ohm
J901
USB-C
J902
U901
MT3608
D902
SS34
U902
X901
25 MHz
U903
74LVC1G17
F901
2A
D901
SMAJ5.0A
LED901
LED902
R901
5.1k
R902
5.1k
R903
1k
R904
12k
R905
1k
R906
51R
R907
1k
L901
22u
C901
10u
C905
1u
C906
1u
C907
100n
+5V
+8V
+3V3
r9_VBUS
r9_CC1
r9_CC2
r9_SW
r9_FB
r9_OSC
r9_BUF
r9_REF
r9_LEDY
r9_LEDG

EXPECT: +5 V from the USB-C (up to 1.5 A), +7.8 V at 250 mA for the gain blocks (yellow LED), a 25 MHz square wave on REF OUT
USB-C power only: 5.1 k on each CC asks any source for 5 V; a 2 A PTC, a 5 V TVS and 20 uF. The MT3608 boosts to 7.8 V (0.6 V x 13 from 12 k / 1 k) for the GALI-84+ tile, r12 (100 mA); the +8V rail is a strip under row 1 that their headers reach with a via. The TPS7A2033 makes 3.3 V for the TC32H4 TCXO (+-2 ppm) and the 74LVC1G17 that drives REF OUT (51 ohm: 1.6 Vpp into 50 ohm, 3.3 Vpp into a synthesizer's unterminated REF IN; r10's REF IN passes it to r13).
The TCXO's pad 1 (enable or no connection on the drawing) floats, as CostasLink's review settled.

r7 TX mixer, ADL5350

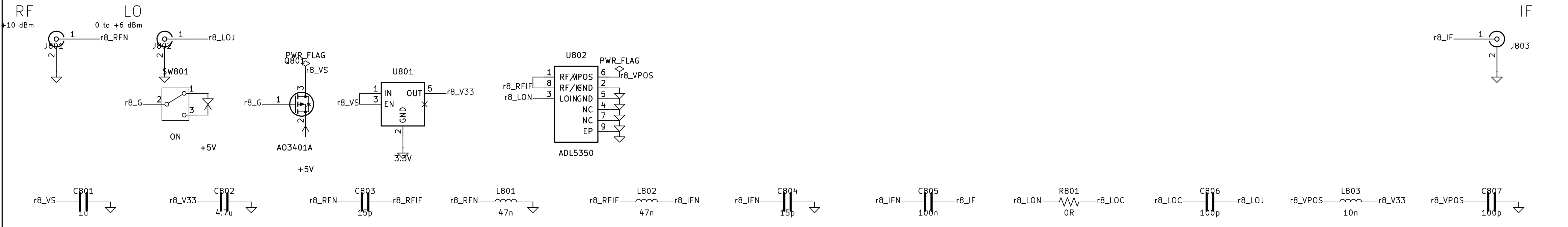
cells (0,2), (1,2)


RF
+10 dBm
LO
0 to +6 dBm
IF
J701
J702
J703
SW701
ON
PWR_FLAG
Q701
AO3401A
U701
U702
ADL5350
+5V
3V3
r7_RFN
r7_LOJ
r7_IF
r7_VS
r7_V33
r7_VPOS
L701
47n
L702
47n
R701
0R
L703
10n

EXPECT: RF x LO to IF (or IF x LO to RF) with 7-9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1-30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band-tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3-8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up-converts a DAC tone (IF in, RF out) or down-converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with r2 or r4.

r8 RX mixer, ADL5350

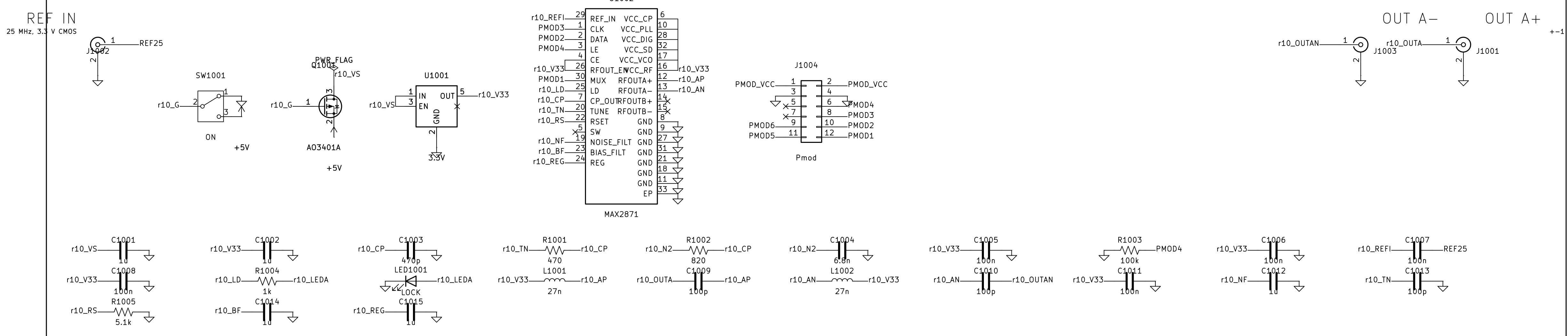
cells (2,2), (3,2)


RF
+10 dBm
LO
0 to +6 dBm
IF
J801
J802
J803
SW801
ON
PWR_FLAG
Q801
AO3401A
U801
U802
ADL5350
+5V
3V3
r8_RFN
r8_LOJ
r8_IF
r8_VS
r8_V33
r8_VPOS
L801
47n
L802
47n
R801
0R
L803
10n

EXPECT: RF x LO to IF (or IF x LO to RF) with 7-9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1-30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band-tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3-8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up-converts a DAC tone (IF in, RF out) or down-converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with r2 or r4.

r10 synthesizer, MAX2871

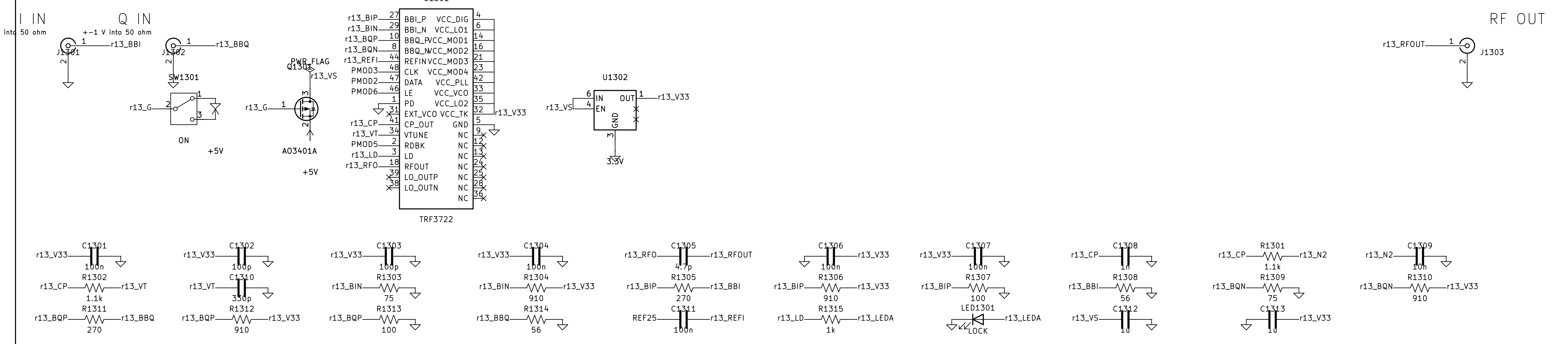
cells (0,3), (1,3)


REF IN
25 MHz, 3.3 V CMOS
OUT A−
OUT A+
J1001
J1002
J1003
J1004
Pmod
SW1001
ON
PWR_FLAG
Q1001
AO3401A
U1001
U1002
MAX2871
+5V
3V3
REF25
r10_OUTA
r10_OUTAN
C1001
1u
C1002
1u
C1003
470p
R1001
470
R1002
820
C1008
100n
R1004
1k
LED1001
LOCK
L1001
27n
C1009
100p
R1005
5.1k
C1014
1u
C1015
1u

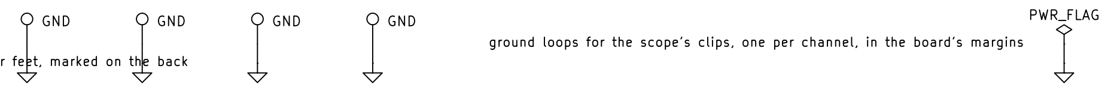
EXPECT: any frequency from 23.5 MHz to 6 GHz on OUT A+ and OUT A− (each −4 to +5 dBm in steps, 180 degrees apart), locked to the 25 MHz on REF IN; the red LED says locked
MAX2871 as CostasLink built it: fractional-N PLL with four VCOs, programmed over SPI from the Pmod (SCLK, SDATA, LE; MUXOUT back). The loop filter is third order, 80-120 kHz with 61 degrees of margin for both 915 MHz (VCO / 4) and 2.4 GHz (VCO / 2) at ICP 2.56 mA (RSET 5.1 k, CP = 7). RFOUTA+ and RFOUTA− each feed a jack through 27 nH to 3.3 V and 100 pF: two outputs, 180 degrees apart; put a 50 ohm terminator on one that is unused. RFOUTB is open and must stay disabled. CE and RFOUT_EN are tied high, LE has 100 k down. REF IN is unterminated; a back trace takes it on to r13.

r13 I/Q transmitter, TRF3722

cells (2,3), (3,3)


I IN
+−1 V into 50 ohm
Q IN
RF OUT
J1301
J1302
J1303
SW1301
ON
PWR_FLAG
Q1301
AO3401A
U1301
TRF3722
U1302
+5V
3V3
r13_BBI
r13_BBQ
r13_RFOUT
R1302
1.1k
C1310
330p
R1303
75
R1304
910
R1311
270
R1312
910
R1313
100
R1314
56
LED1301
LOCK

EXPECT: I + jQ from two DACs on 0.4-4.2 GHz, about −3 dBm, image-free (−42 dBc untrimmed)
TRF3722 (PLL, VCO, I/Q modulator) on SPI from the Pmod (LE IO6, readback IO5); reference from r10's REF IN. Each DAC: 56 ohm at the jack, 270 ohm to a node held at 0.25 V (+-0.25 V for +-1 V). DC on I alone is a CW carrier.


GND
GND
GND
GND
PWR_FLAG

no mounting holes: four stick-on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/rf_mod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a quadrature transmitter, a synthesizer, three mixers (one a diode ring), an LNA and a gain block, two
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/
Sheet: /
File: rf_mod.kicad_sch
Title: RF-MOD: 915 MHz, 2.4 GHz and a stretch to 5.8 GHz
Size: A1
Date: 2026-10-25
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1